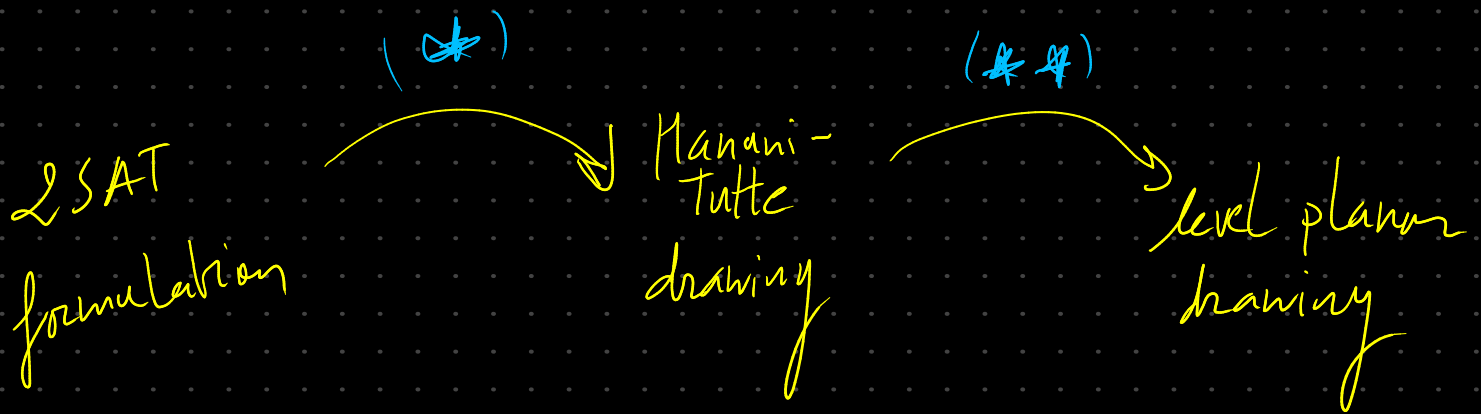


Summary of my understanding



(**)

Th2:

G has
an $\#$ -monotone
drawing
with every independent
edges crosses
evenly

$\Rightarrow$

$\#$ -monotone
embedding in
which each
vertex keeps
its $\#$ -coordinates

(*)

Th1

$S(G)$ is satisfiable $\Leftrightarrow G^*$ has a Hanani-Tutte level drawing

so we get to the next basic level of understanding

Th₁: $S(w)$ is satisfiable $\Leftrightarrow w^*$ has a Hanani-Tutte level drawing

Th₂:

w has an x -monotone drawing with every independent edge crosses evenly

$\Rightarrow$

x -monotone embedding in which each vertex keeps its x -coordinates

the next question is: can we construct a level planar drawing from the 2SAT formulation? (this is the Heris question)

$\rightarrow$ how about check this implication
path $(*) \Rightarrow (**)$

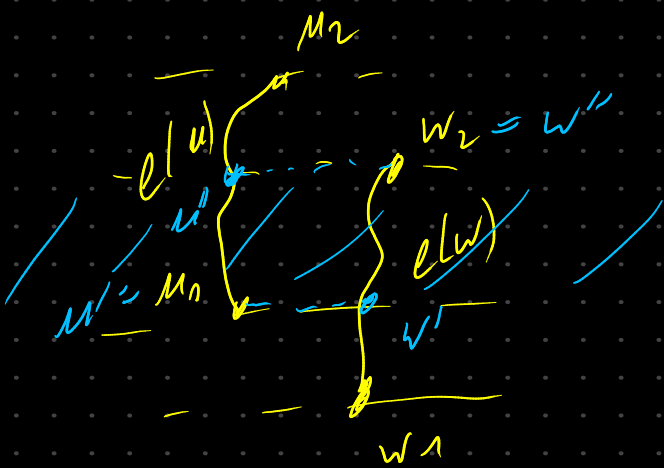
so first subquestion

2SAT $\longrightarrow$ H-T Drawing

construction of a satisfying truth ass. of $S(h)$
 from a given HT $\perp$
 more interested in lemma 4

satis. t.a. φ^+ of $S(h^+)$ from $\varphi \upharpoonright_{V(h)}$
 input φ satisfying $S(h)$

$u, w \in V(h)$ with $l(u) = l(w)$



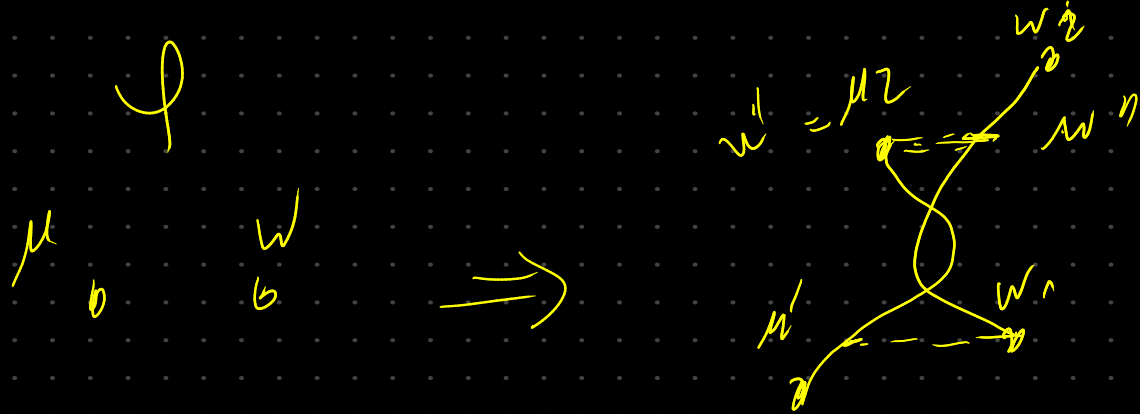
if we set

$$\varphi(u, w) = \varphi^+(u', w')$$

because φ^+ satisfies the clause so

$$\varphi^+(u', w) = \varphi^+(u'', w'') \Rightarrow \varphi(u, w) = \varphi(u'', w'')$$

if for $u, w \in h$ / $l(u) = l(w)$
 have $\varphi(u, w)$

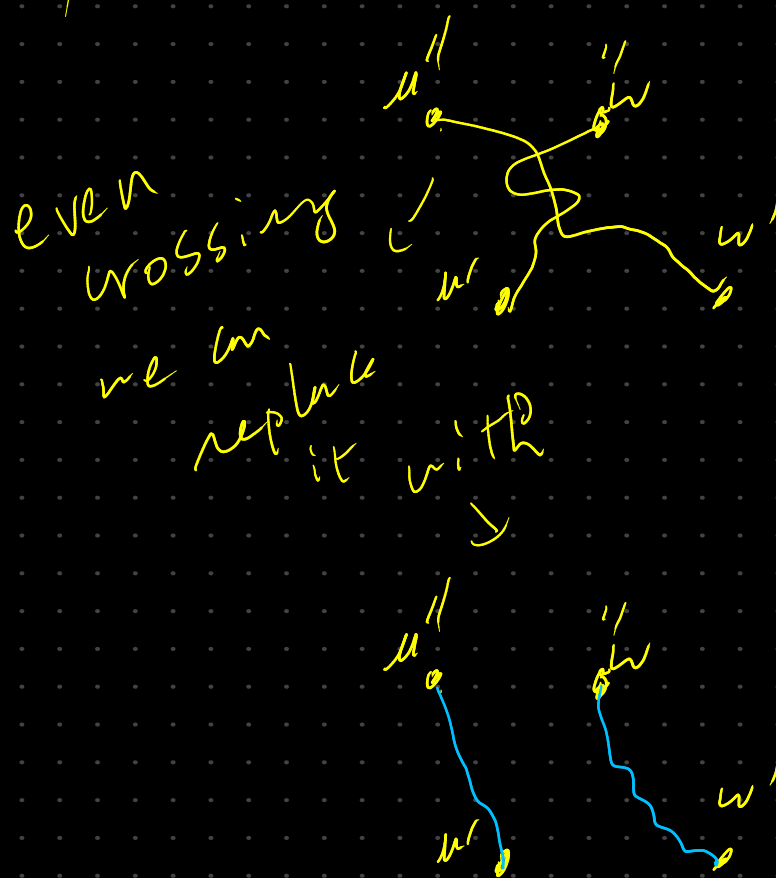


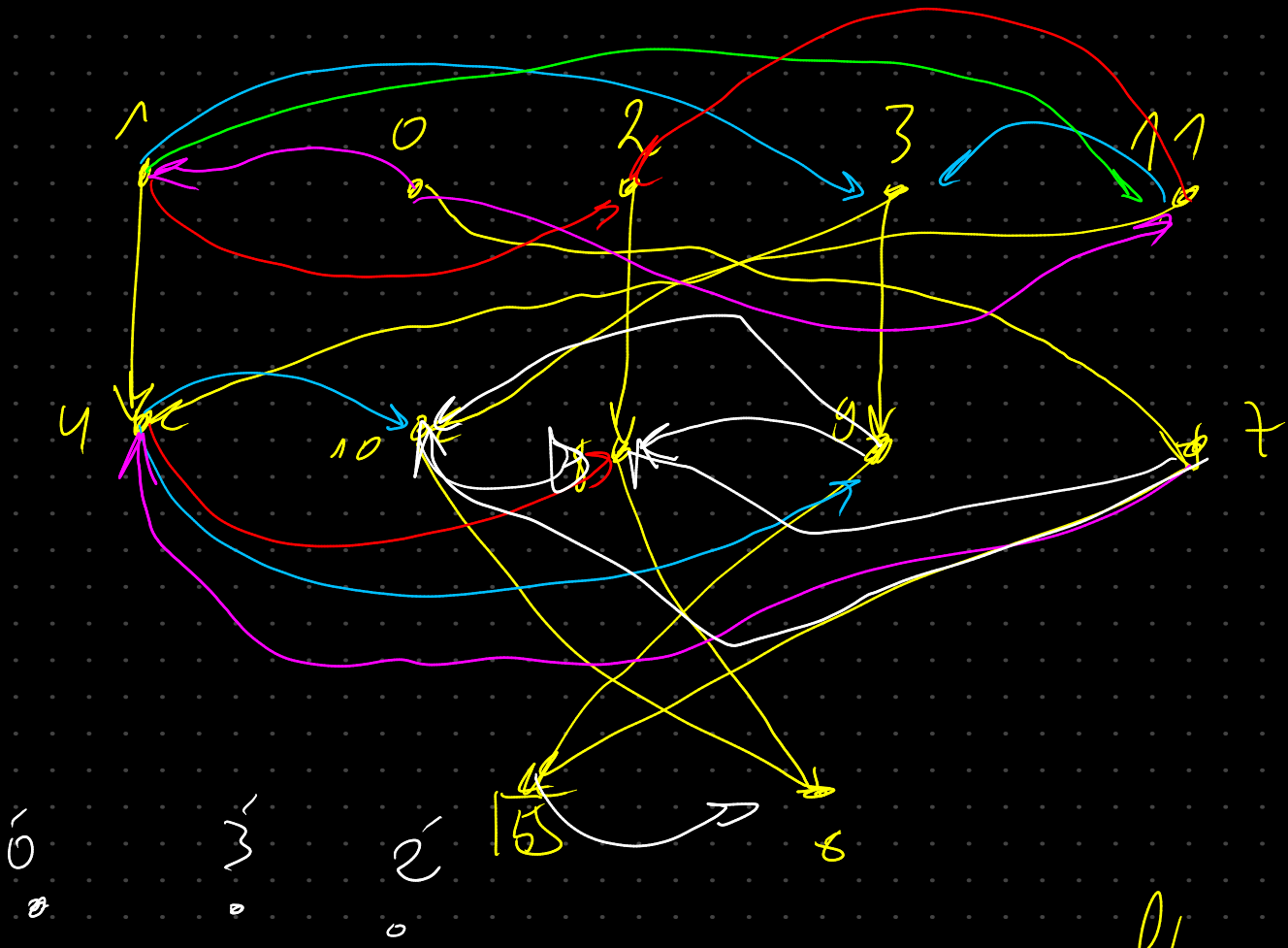
set $\varphi^+(\mu w) = \mu_1$

because φ^+ is a set of $S(\mu^+)$: $\varphi^+(\mu'' w'') = \varphi(\mu w)$

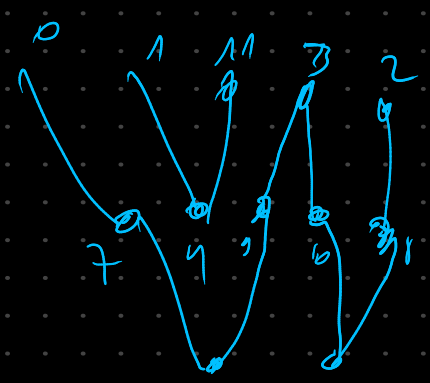
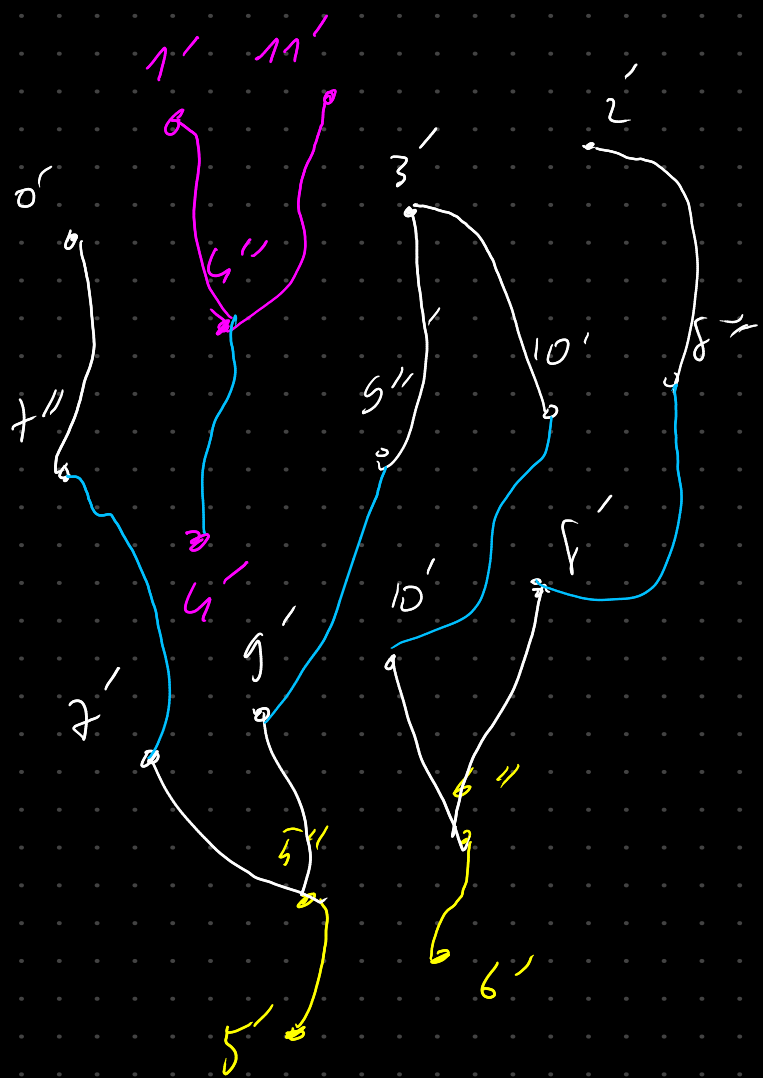
same is true for all subdivision vertices

in $e(u)$, $e(w)$ in ht





$$f(xy) = f^+(x'y')$$



we start from first level

$$\varphi(5,6) = \text{true}$$

$$\Rightarrow \varphi^+(5', 6') = \varphi^+(5'', 6'') = \text{true}$$

l_1

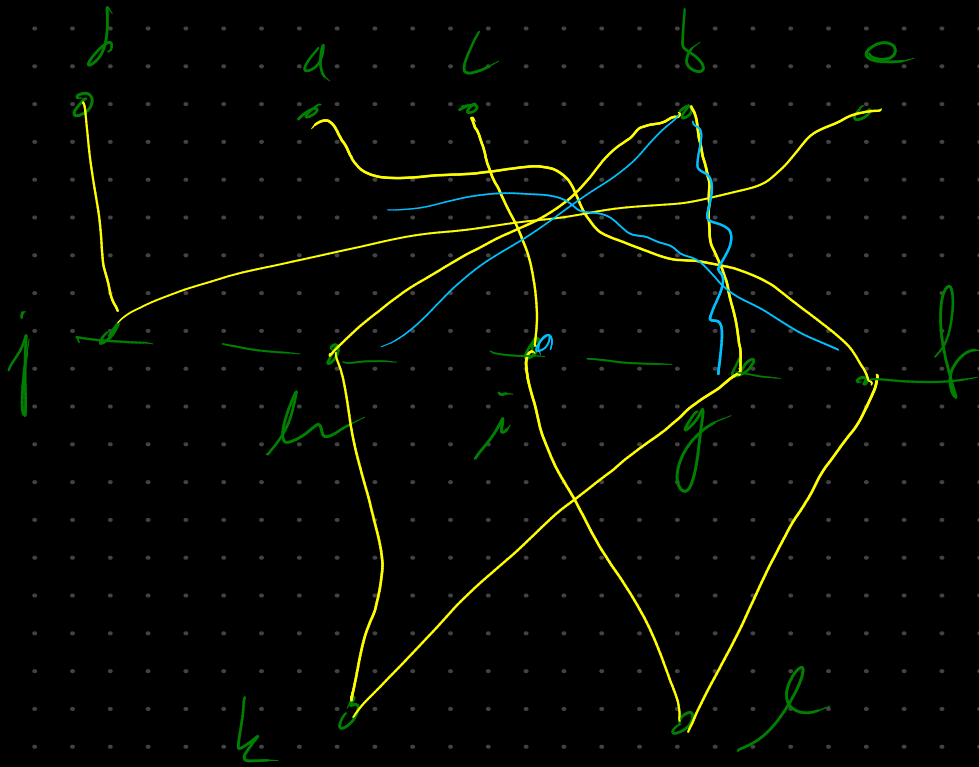
$$p(4,10) = 1, \quad p(4,8) = 1, \quad p(9,9) = 1$$

$$f(4,7) = 0$$

$$f(10, 8) = 1, f(10, 9) = 0, f(10, 7) = 6$$

$$f(8,9)=0, \quad f(8,7)=0,$$

$$\varphi(y, \tau) = 0$$



$$\begin{aligned} & \varphi(ab) \\ &= \neg \varphi(hf) \\ &= \neg \neg \varphi(hg) \\ &= \varphi(hg) \end{aligned}$$

$$\begin{cases} \varphi(ad) = \varphi(ij) \\ \varphi(ac) = \varphi(ij) \end{cases}$$

$$\begin{cases} \varphi(ij) = \varphi(cd) \\ \varphi(ij) = \varphi(ce) \end{cases}$$

$$a < d < b \Rightarrow a < b$$

$$k < l \Rightarrow \underline{\underline{g < i}}$$

$$i < j < g \Rightarrow i < g$$

$$i < g \Rightarrow k < l$$

$$i < j < g \Rightarrow g < i$$

$$\begin{aligned} a < b \\ \varphi(ad) &= \varphi(ij) \\ \varphi(ab) &= \varphi(jg) = \varphi(ih) \end{aligned}$$

Coming back again to Rutter paper
there is a tradeoff of transitivity
w/ crossing

but also
not considering
crossing
change.

$$\varphi \text{ of } S(w) \Rightarrow \varphi^+ \text{ of } S(w^+)$$

Γ^+ of h^+ induced by Γ^* and
 Ψ^+ t.a. induced by Γ^+ (not nec. the crossing class)

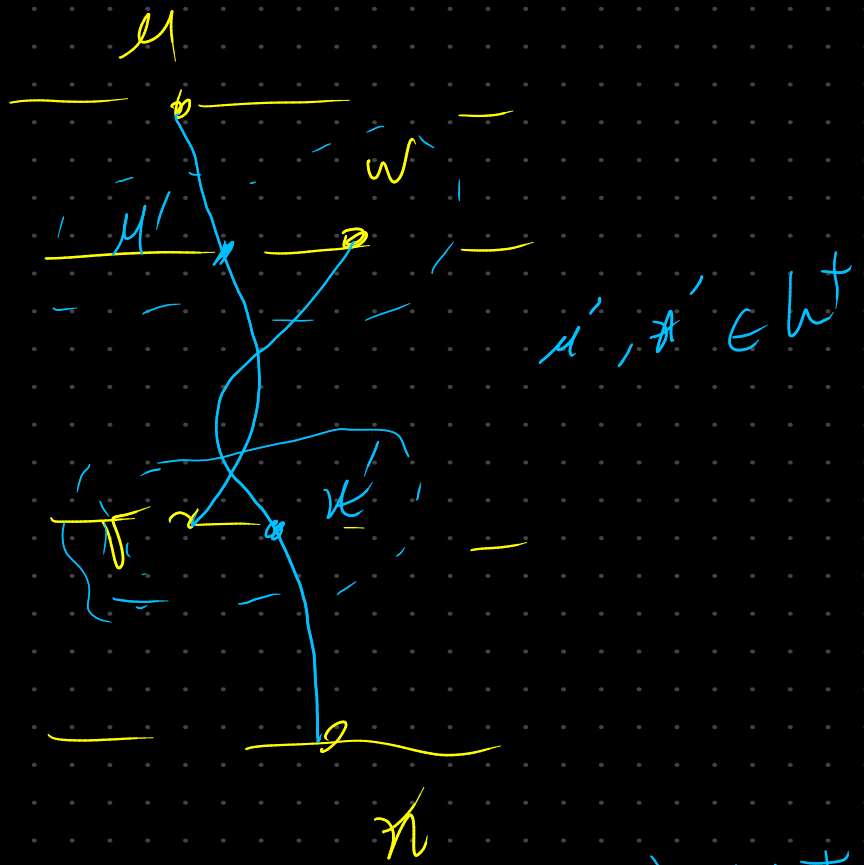
Ψ^+ t.a. induct by Γ^+ (rel. the crossing class)

$$u'', w'' \in V(h^+) \text{ with } Q(u'') = Q(w'')$$

(1)

$$M \in V(W^*)$$
$$R \quad W'' \in V(W^*)$$
$$u'', r'' \notin V(h^*)$$
$$\begin{aligned} \psi^+ (w' w^n) \\ = \psi^+ (w'' w'') \end{aligned}$$

because
transitivity is
satisfied
in $S'(h^+)$



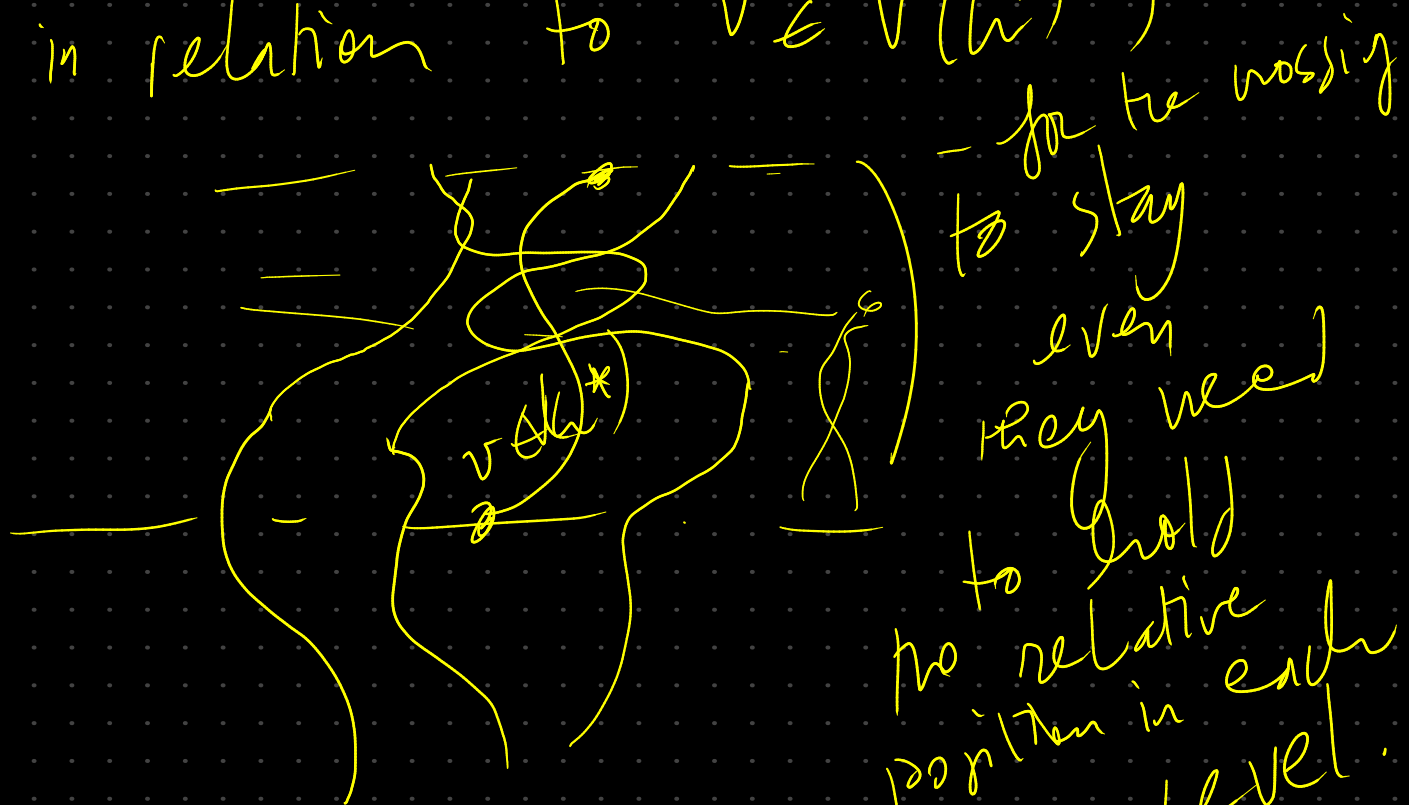
$((u, v), (w, x) \text{ intersect event}) \quad HT$

$$\Rightarrow \varphi^+(x' w') = \varphi^+(v' x')$$

I think the idea is that H-T have one vertex per level so the transitivity clause is always satisfied.

the clause is satisfied for h^* if we induce h^* with more vertices it doesn't change the fact we just annotated more.

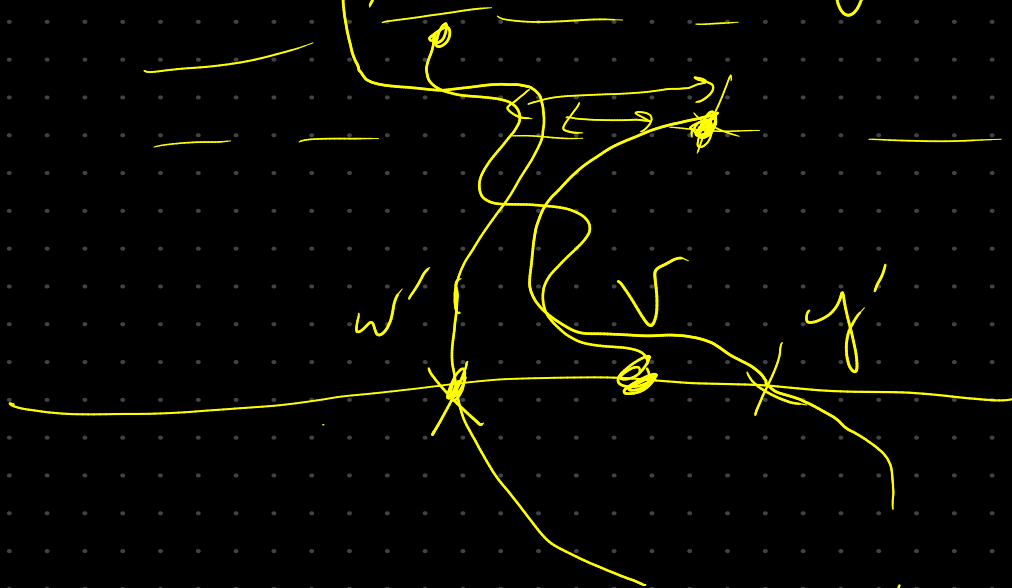
(when it come to edge position in relation to $v \in V(h^*)$)



I think that's why transitivity clause hold for h^* and h^* induced by h^*

Critical edges are "sorted" in the critical area, so +

Lemma 2 is kind of a proof that
 hammett ^{hans} respect transitivity does



the relative order of curves to a
 vertex on level i is the same
 as the relative order it was in the
 last level of intersection. $uw \wedge$
 $wy \Rightarrow$
 uw



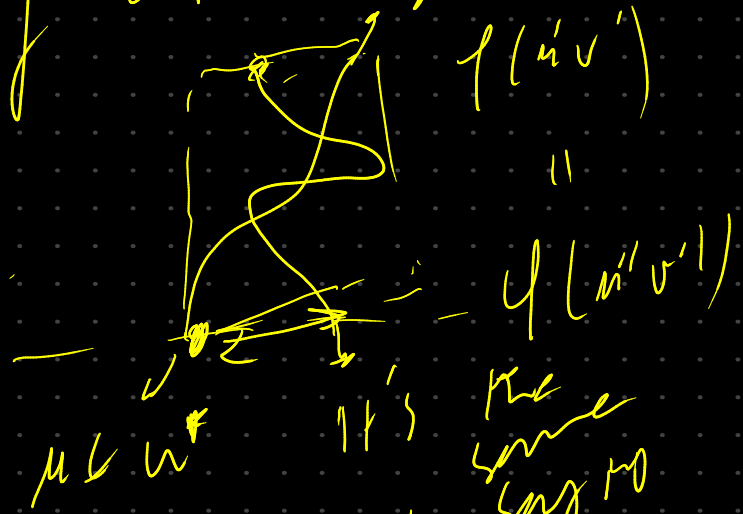
ψ^+ t.a. induced $\in \Gamma^+$

f^+ satisfies
all laws
of $S(h^+)$
(1)
(2)

$u'', w'' \in V(h^+)$

if $w'' \in V(h^+)$ w.l.g.

if we are on H- τ drawing
even crossing of critical edges.
it means that



Let
 $\psi^+(u''w'')$

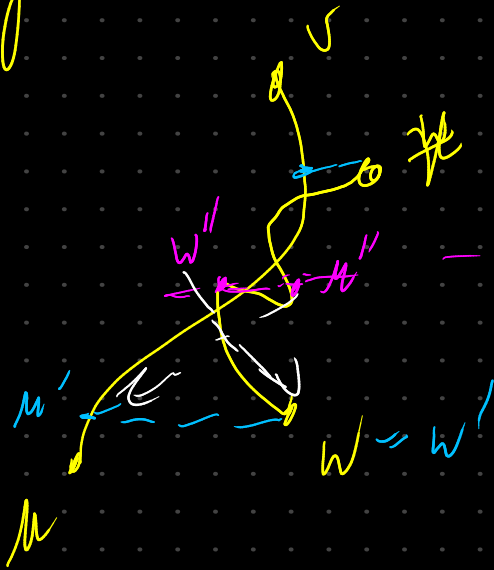
$= \psi^+(u''w'')$

if we want to satisfy
the crossing
between the vertex
in h^+ and another vertex

that satisfies
the crossing
constraint

$f(u'v') = f(u''v'')$
and
also no crossing

the issue would be between
to subdivision nodes
of $(u, v), (w, x) \in E(h^*)$



$$\varphi^+(u''w'') = \varphi^+(u'w')$$

$\Rightarrow$

$$\varphi^+(u''w'') = \varphi^+(u'w')$$

$$\varphi^+(u''w'') = \varphi^+(u'w')$$

$$\Rightarrow \varphi^+(u''w'') = \varphi^+(v''x'')$$

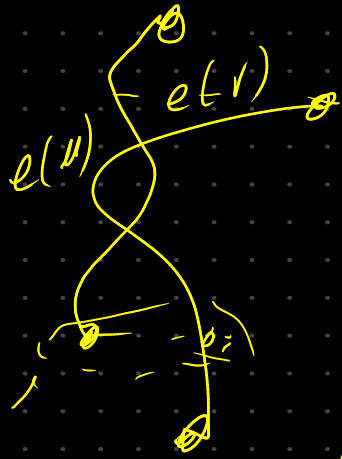
$\varphi(u'w'), \varphi(v''x'') \notin E(h^*)$
with identical

hence satisfies $S(h^*)$, how to
 $S(w)$? $u, v \in V(h)$ with $l(u) = l(v)$

for each level in h^* we have planarity constraint.

$\Rightarrow$

we have stretched edges $e(u)$, $e(w)$
 in h^* are critical there induced
 representation in h^* would be



which falls
 in the category
 we talked about
 in 1)

meaning that $\varphi^+(e'v') = \varphi^+(u''v'')$



we can synthesize it to 2 points
 in a level with no crossings



φ^+ satisfies

and φ satisfies too.

$$\begin{aligned} & \text{see } \varphi(uv) \\ &= \varphi^+(uvv') \\ &= \varphi^+(u''v'') \end{aligned}$$

$$(u', u''), (w', w'') \text{ lin of } e(u), e(w)$$

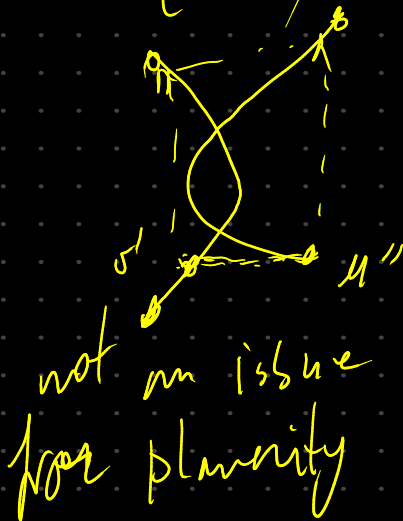
first we want to map it to $S(u^+)$
 we call ψ^+ satisfied t.n in $S(u^+)$
 the issue (planarity constraint)

consider

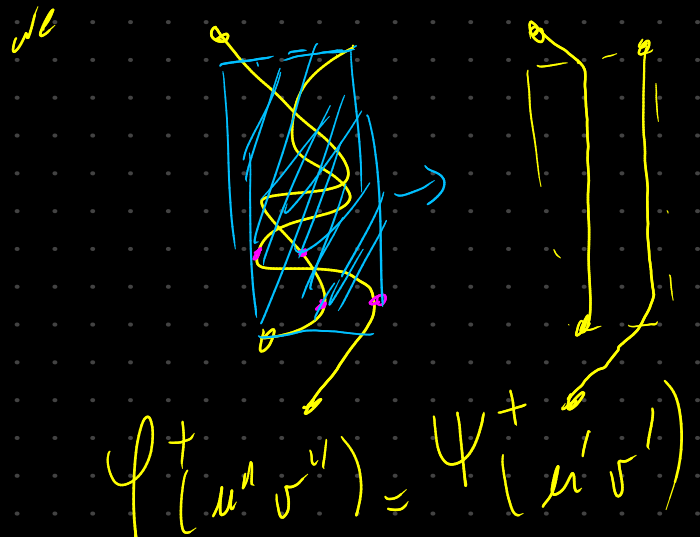
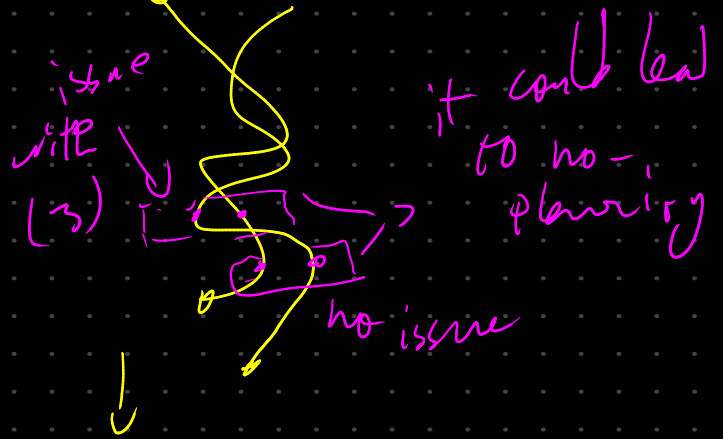
$$u', u'' \in V(u^+)$$

$$v'' \notin u^+ \\ u'' \in u^+ \\ \text{w.l.g.,}$$

$$\psi^+(u''v'') \\ = \psi^+(u''v')$$



$$v', u'' \notin u^+$$



there by we have $\varphi^+(u'' w'') = \varphi^+(v'' x'')$
 for any two critical edges. (planarity)
 and clearly $\varphi^+(u w) = \neg \varphi^+(w u)$
 so φ^+ satisfies $S(h^+)$

now we want to link φ^+ to φ

$\downarrow$
 $S(h^+)$

$\downarrow$
 $S(h)$

first we focus on the vertices of
 the same level, can we map relative
 order of some vertices ^{according to} $S(h^+)$ to $S(h)$?

so we start with the stretch edge
 $u, v \in V(h)$



$\varphi^+(v' u') = \varphi^+(v'' u'')$
 every vertex in the
 critical area has
 the same order so

let $\varphi(u w) = \varphi^+(u' w')$ ^{planar} $\Rightarrow \varphi(u w) = \varphi^+(u'' w'')$

meaning that $\ell(uv) = \ell(wu)$

as all the vertices in $e(u)$, $e(w)$ has the same name

if there is any crossing^{oth} between
stretch edges it means there
is an issue between making 3

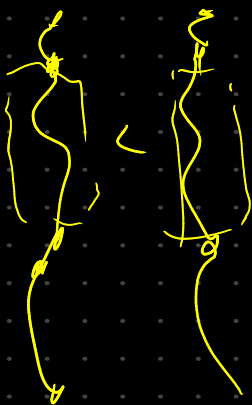
levels

$$u \equiv e(u)^{u+}$$

$$v \equiv e(v)^{v+}$$

$$\rightarrow \text{if } \left. \begin{matrix} e(u) \\ \text{in } u^+ \end{matrix} \right\} < \left\{ \begin{matrix} e(v) \\ \text{in } v^+ \end{matrix} \right\} \Rightarrow u < v$$

consistency (because there is no crossing)



from $s(u)$ to $H-T$ (homotopy)

we have $S(u) \rightarrow S(u^+) \rightarrow H-T$

drop
some
of the
crossing
constraint

coming back to the proof of Lemma 3:

Let $u \in V(u^+)$ and $v \in V(u^+)$



because Γ is $H-T$
drawing

$\Downarrow$
critical edges cross
evenly

$\Downarrow$
can only cross
in the critical
area

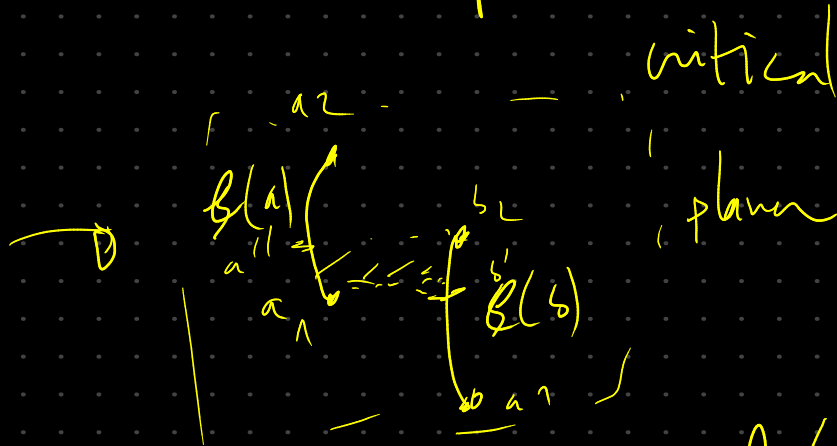
$\Downarrow$
 $\psi^+(v, u)$
 $= \psi^+(v, u)$

So for Lemma 4 ($S(h) \rightarrow HT$)

t.a.p satisfies $S(h)$

let p^+ t.a. that satisfies $S(h^+)$

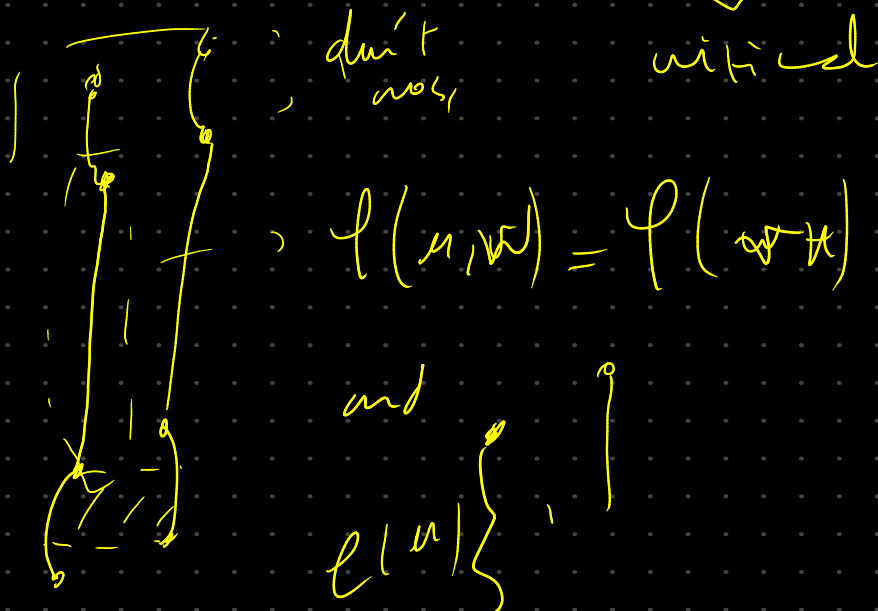
$a, b \in V(h)$
with
 $\ell(a) = \ell(b)$



we have
 $\varphi \rightarrow$ planar ext.
 $\downarrow$ consistency ext.

so $\varphi^+(a_1, b_1) = \varphi(a, b)$
as every part of the critical
area we just need to

prove that also φ respect planarity for
edges representing $(u, v), (w, x)$ in h



$$\ell(u, v) = \ell(w, x)$$

and

$$\ell(u) = \ell(w)$$

so we just prove $\varphi(u'v') = \varphi(uv)$

$$\varphi(uv) = \varphi(w^*)$$

$$\varphi^+(u'v') = \varphi^+(w^*)$$

$$\varphi^{+''}(u'v'')$$

start point

L^+ (a planar drawing)

$S(L^+)$ consistency est
planarity est

based on pt of L^+

~~est~~

let $u', v' \in V(L^+)$

$\downarrow$
 $u', v' \in e(u), e(v) \quad \beta \quad u, v \in V(L)$

$$L(u) = L(v)$$

u', v'

part of $e(u), e(v)$

part of
edges
connecting
vertices not
rep. v- from
in L^+

$$\varphi^+(u'v') = \varphi(uv)$$

$$\rightarrow \varphi^+(u'v') = \varphi(w^*)$$

we get a tree satisfying $S(L^+)$

in every level there is at least
one vertex that is not subdiviser vertex.

Let Γ denote it as a sub-div.

Let Γ^* drawing of h^* induced by Γ^+ .

$(u, v), (w, x)$ critical edges. of h^*

$(u', v'), (w', x')$ denote their limits in h^+ .
one vertex of each limit on the two levels is a subdivided vertex.

By lemma 2: $\varphi^+(u', w') = \varphi^+(v', x')$

$\Rightarrow (u, v)$ and (w, x) cross each other

in Γ^*



by lemma 1 we have φ^+ is a topological drawing

so if u, v, w, x are critical edges
then $\varphi^+(u', v') = \varphi^+(w', x')$

with lemma 1. This results in a drawing with even crossings between (u, v) and (w, x)